

DDTA BA4-T3 projection closure review - R1

Status: CLOSED / PASS

Repository baseline reviewed: dcb4605448de4ac5331f10ff090a9f2ab677427e

Executed scope: BA4-T3 only - integrated projection boundary, interpretation/coverage and cross-projection closure review.

1. Closure question

Can the combined BA4-T1/T2 candidate be reduced further while still preserving authority, semantic rendering, coverage/completeness, qualification, method-rule accountability, cross-projection comparison and change rebuild?

The review attempts to falsify the candidate by deleting or merging responsibilities before accepting closure.

2. Inputs

The review uses:

- BA4_PROJECTION_BOUNDARY_TRACEABILITY_CANDIDATE_R1.md;
- BA4_T1_PROJECTION_BOUNDARY_TRACEABILITY_SEMANTIC_PRESERVATION_R1.md;
- BA4_PROJECTION_BOUNDARY_INTERPRETATION_COVERAGE_CANDIDATE_R2.md;
- BA4_T2_METHOD_INTERPRETATION_COVERAGE_CROSS_PROJECTION_R1.md;
- DDTA_CURRENT_RESEARCH_STATE_R15.md;
- closed BA1, BA2 and BA3 contracts; and
- the facial M1-M4, order/WMS and provider-normalization controls already used by BA4-T1/T2.

No complete threat-method schema, AnalysisRecord/Common Finding or ThreatForge implementation model is introduced.

3. Attack A - remove coverageMode

Candidate simplification:

```
selectionCoverageContract:  
  "current transfer propositions"
```

with no explicit statement of selective vs exhaustive intent.

Two valid consumers can read that same scope as:

```
A: show every current transfer  
B: show only transfers useful to one analysis question
```

The omission of one eligible transfer therefore cannot be classified as either a defect or an intentional omission.

Disposition: removal **REJECTED**. coverageMode remains semantically required.

The mode does not require one serialization; an unambiguous textual equivalent is sufficient.

4. Attack B - retain omissionSemantics as an independent field

T2 carried both coverageMode and omissionSemantics.

T3 asks whether the latter adds independent necessary semantics.

Once mode semantics are fixed:

EXHAUSTIVE_FOR_DECLARED_SCOPE

omitted eligible+qualified item -> projection defect

SELECTIVE

omitted eligible item -> no project-semantic conclusion

an independent omission policy can only duplicate those rules or contradict them.

Disposition: omissionSemantics as an independent mandatory field **REJECTED AS REDUNDANT**.

This is the only material simplification accepted by the closure review.

5. Attack C - collapse qualificationPolicy into generic coverage text

A scope such as `transfer propositions` does not state whether stale, diagnostic or pending material is admissible.

A current-state projection and a review projection can have the same semantic topic but intentionally different admissible BA state.

If the distinction is hidden only in rendering conventions, a non-visual consumer can ingest stale/unresolved basis as if current.

Disposition: semantic collapse **REJECTED**.

`qualificationPolicy` may be physically nested in coverage metadata, but its responsibility remains distinct.

6. Attack D - remove interpretationRuleRef

The T2 overlapping-rule counterexample is replayed:

R1: `serviceUse + responsibilityBoundary`

-> `external-service exposure`

R2: `serviceUse + responsibilityBoundary + constraints`

-> `assurance dependency review`

An item with only a BA trace does not identify which local rule produced its meaning when several rules overlap.

Disposition: removal **REJECTED**.

`interpretationRuleRef` remains required for meaning-bearing method-owned interpretation. It remains projection-local and never becomes a BA3 derivation rule.

7. Attack E - make trace roles mandatory everywhere

A single BA element mapped directly to one human bullet does not need an artificial contribution role merely to satisfy serialization uniformity.

However, role distinctions are essential when multiple BA inputs contribute differently to one interpretation.

Disposition: universal mandatory role key **REJECTED**; conditional role binding **ACCEPTED**.

Final rule:

traceRoleKey required where contribution roles differ

8. Attack F - collapse shared rendering and method-owned interpretation

F1 semantic-preserving aggregate

C constraint

I constraint

provenance constraint

-> enumerated "delivery constraints" group

The group is no stronger than its BA basis and can remain shared rendering.

F2 consumer conclusion

same constraint basis

-> "high-assurance channel"

This is a stronger consumer interpretation. If admissible at all, it requires a method-owned rule and cannot be presented as shared BA meaning.

Disposition: collapse **REJECTED**.

The semantic distinction remains even if one physical item contains both a shared rendering and local annotations.

9. Attack G - force one shared projection ontology

T3 replays three projections over the same BA:

1. a human project-understanding view;
2. a flow/exposure method view; and
3. an assurance/responsibility method view.

The method views use intentionally incompatible local categories. Both can remain valid because their common denominator is role-bound BA trace, not category equivalence.

The human view can group the same BA basis without adopting either method taxonomy.

Disposition: universal projection ontology **REJECTED**.

No SharedProjectionConcept or method-category equivalence relation is forced.

10. Attack H - require cross-baseline projection-item identity

M1-M4 are replayed under rebuild semantics.

A local projection item at B0 may disappear, split or be represented differently at B1 because the BA meaning or the projection descriptor changed.

The project-semantic continuity already exists in BA3. Adding RETAIN | REPLACE | RETIRE to projection items duplicates authority without current evidence.

Disposition: projection-item BA-like lifecycle **REJECTED AS UNNECESSARY**.

Method/UI continuity remains tool-owned if desired.

11. Integrated rebuild regression

M1 Ethernet -> Wi-Fi

Flow projection may remain materially stable while realization/assurance projection changes. PASS: different projection deltas are expected because coverage differs.

M2 external -> project-owned transport

The external-service exposure interpretation becomes non-applicable when the BA responsibility/service basis changes. A different method can produce a project-owned authority item. PASS: no common method category is required.

M3 remote -> local recognition

Retired remote transfer disappears from a current exhaustive flow view. Unrelated responsibility items can remain. PASS: rebuild follows BA rather than local item history.

M4 raw capture -> opaque reference conflict

A current projection excludes stale/unresolved transfer meaning; a review projection may surface it with qualification. PASS: no projection invents post-resolution project truth.

12. Order/WMS regression

Externalizing inventory authority can produce large deltas in an integration projection and different deltas in a responsibility/assurance projection.

Each remains consistent if it follows its own declared scope, mode and qualification policy and traces to accepted BA.

No method-local `ExternalSystem`, `TrustBoundary` or `IntegrationRisk` category is forced into BA.

Disposition: PASS.

13. Provider-normalization regression

A consumer needing `PaymentAuthorizationResult` must consume accepted normalized BA meaning.

The negative pattern remains rejected:

```
provider raw vocabulary
  -> hidden method mapping
    -> supposedly shared project result
```

The accepted path remains:

```
provider raw vocabulary
  -> governed mapping / BA3 derivation
    -> accepted BA meaning
      -> projection interpretation
```

Disposition: PASS.

14. Orthogonality check

The closure review confirms four independent questions:

trace

Which BA meaning supports this projection item?

coverage

Is the view selective or exhaustive for its declared eligible scope?

qualification

Which BA states are admissible, and how is non-current meaning preserved?

interpretation rule

Which consumer-owned rule justifies this non-trivial local meaning?

No pair can be merged without losing a reviewed distinction.

Physical co-location remains allowed.

15. Earlier-layer reopen checks

No material counterexample forces an earlier reopen.

BA1: BAreferent + BAProposition remain sufficient PASS

BA2: current operator/role vocabulary remains sufficient PASS

BA3: provenance/lifecycle/change continuity sufficient PASS

A method-specific node, edge, risk class, assurance class, UI state or local identifier is not a shared-semantic counterexample.

16. Closure decision

BA4-T3 CLOSED / PASS

BA4 CLOSED FOR CURRENT THESIS SCOPE

The accepted final contract is:

- BA4_PROJECTION_BOUNDARY_TRACEABILITY_INTERPRETATION_COVERAGE_CONTRACT_R1.md

BA4-T1/T2 artifacts remain immutable research derivation history.

17. Closure ledger summary

projection authority	REJECTED
immutable projection revision	RETAIN
eligibleBAScope	RETAIN
coverageMode	RETAIN
independent omissionSemantics	REMOVE / REDUNDANT
qualificationPolicy	RETAIN
BA trace	RETAIN
traceRoleKey mandatory everywhere	REJECT
traceRoleKey where roles differ	RETAIN
shared semantic rendering	RETAIN
method-owned interpretation	RETAIN
interpretationRuleRef	RETAIN WHEN APPLICABLE
universal projection DSL	REJECT
universal projection ontology	REJECT

cross-projection shared identity	NOT FORCED
BA trace + BA3 continuity common anchor	ACCEPT
projection lifecycle ontology	REJECT
new BAE family	NOT FORCED
new BA2 operator	NOT FORCED
BA1/BA2/BA3 reopen	NOT TRIGGERED

18. BA4 reopen criteria

Reopen only the smallest responsibility on a material case where the final contract cannot express necessary projection completeness, cannot preserve qualification, cannot reproduce method-owned interpretation, loses source drill-down, requires a shared semantic identity beyond BA trace, or cannot rebuild from BA3 continuity without a second shared lifecycle.

Do not reopen for ordinary method taxonomy growth, UI identity needs, renderer changes, new projection revisions or storage preferences.

19. Next phase boundary

BA5 may begin only after this BA4 closure package is committed, pushed and remotely verified.

BA5 must not reinterpret projection-local labels as BA semantic keys. Its first trial must pressure the boundary among source wording, stable BA semantic keys, projection display labels and authoring synonyms before optional assistance is considered.